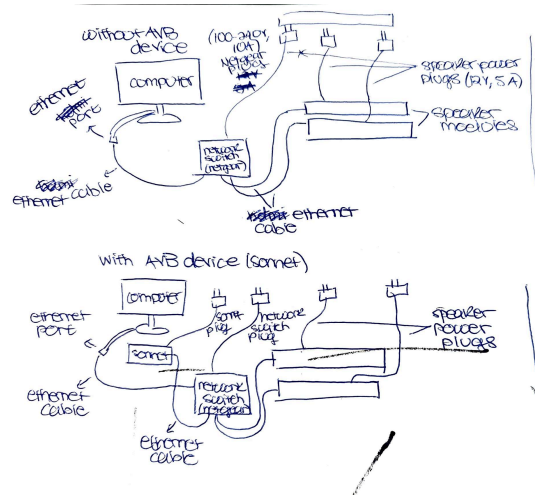


Proj Documentation

Aspects:

Speaker setup diagram:



Speaker setup:

Without AVB Device

Instructions:

- On your computer (mac)
 - Go to Audio MIDI Setup
 - Use (+) in the bottom left corner to create an Aggregate device
 - Select audio modules as part of the Aggregate device
 - Open Reaper and test reaper 1-8 to make sure modules and speakers are connected to computer**

With AVB Device

Instructions:

- On your computer (mac)
 - Make sure that the audio driver that corresponds to the soundcard in the Sonnet is installed/updated. As of 7/2/2026, using the HDSPe AoX-M in the Sonnet Echo Express SE II
 - Go to System Settings > Extensions, and scroll down to Extensions by app. A settings app for the driver as of 7/2/2026 should appear in the Extensions by app list.
 - Make sure "Driver Extension" is switched on
 - Open HIVE (an app by Milan AV)
 - Modules should appear in Hive, as well as the soundcard
 - Change the output stream on the soundcard to be 32 channels for each module
 - [To be continued]

Software (MAX/MSP, Reaper, Spat5, Spat Revolution)

Example patches and scripts can be found in the max patches and reaper scripts folders.

Software for spatial audio - Benefits and disadvantages

There are many different pipelines you can use for spatial audio. The pipelines that I played with during the summer are:

Reaper + Spat Revolution

Reaper is a Digital Audio Workstation (DAW), and Spat Revolution is a software that emerged out of the Spat5 library originally made for MaxMSP. You can use Reaper to send audio to Spat Revolution using VST plugins, which are automatically installed with Spat Revolution.

Benefits

- Can easily send commands to Reaper through Lua/Python/C++ scripting (API calls)—don't need to do anything with mouse

Disadvantages

- Spat Revolution is a paid service (very expensive), and we don't have a subscription
- Spat Revolution is not screen reader compatible
- Spat5 with Max/MSP seems to offer more flexibility & more experimental algorithms

Max/MSP + Spat5 library Use this!

Max or Max/MSP is a visual programming language for multimedia production, most notably for audio. Spat5 is a library developed by IRCAM that can be downloaded off the IRCAM website. It offers sound spatialization algorithms and GUI for moving around sounds in 3D space.

Benefits

- Much cheaper than Spat Revolution (\$60 for Max/MSP vs. 200-2000 dollars for Spat Revolution)
- Able to create Max patches with javascript
- Able to customize speaker and room setup w/in max programming language for Spat

Disadvantages

- Visual programming language—not screen reader compatible
- Not as intuitive as Spat Revolution or even Reaper—new programming language to learn (although quite simple and specific)

Reaper alone

Reaper can route audio up to 512 individual channels, so you can mix audio for large multichannel productions or play audio through very specific speakers.

Benefits:

- Reliably sends audio through specific channels
- Easily scriptable (Python/Lua/C++)
- Free

Disadvantages

- Can't liberally move around sound objects like Spat5 or Spat Revolution; panning algorithms + an interface to move around sound object aren't built into the system

Likely Best:

- Perception experiments where sound needs to play through a specific channel.

Due to each pipeline's advantages and disadvantages, I ended up using Max/MSP + Spat5 library to set up spatial audio through the 8 speaker modules.

System Capabilities of Max:

- Types of spatial audio panning
 - VBAP, DBAP (2D and 3D versions), WFS (unclear what collection of algorithms are being used), Binaural audio, HOA (higher order ambisonics)

- Be mindful that different types of spatial audio panning require different speaker setups (ex. VBAP appears to work best with triangular speaker setups). Spat5 will throw errors I made a short python script that generates speaker coordinates depending on some speaker module specifications; if you want a rough estimate of how your speakers will behave, can use to generate.
- Helpful system allowances
 - Send OSC commands to move around audio in space. Examples of using pythonOSC to send OSC commands are included in the code
 - Upload custom SOFA files to create binaural audio